

Networks

Properties of trees

Michel Bierlaire

Optimization: principles and algorithms

Properties of trees

Trees have some interesting properties

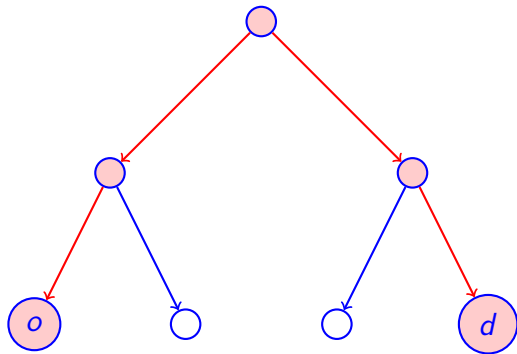
We review some of them.

We also provide some characterizations of tree, involving these properties.

Lemma 21.9

A tree with at least one arc has at least two leafs.

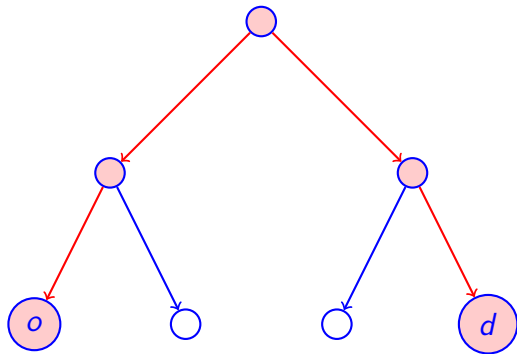
- Path $\mathcal{P}$ with maximum number of arcs.



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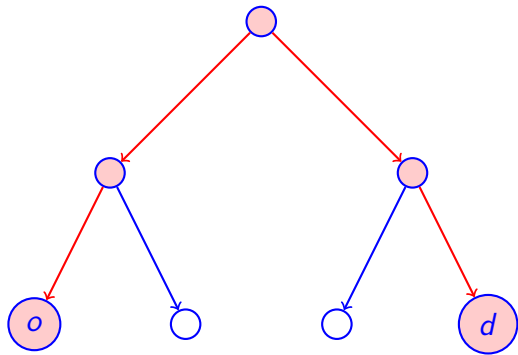
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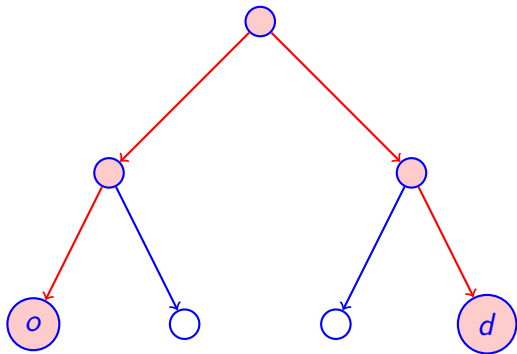
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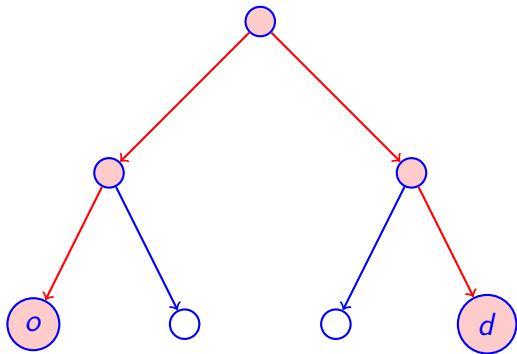
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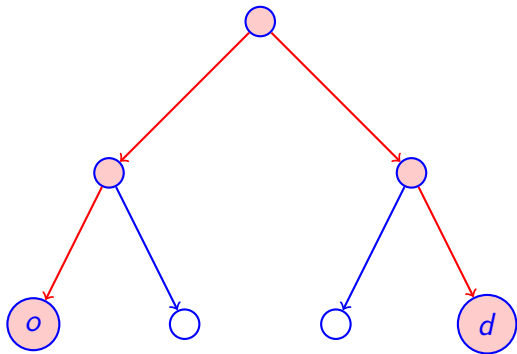
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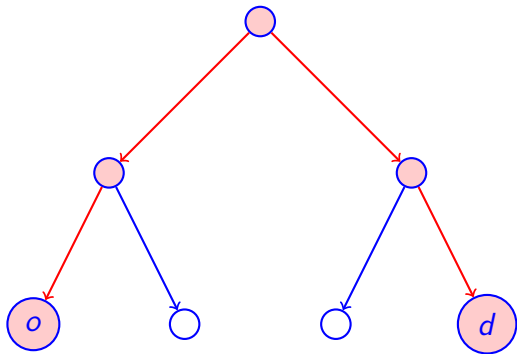
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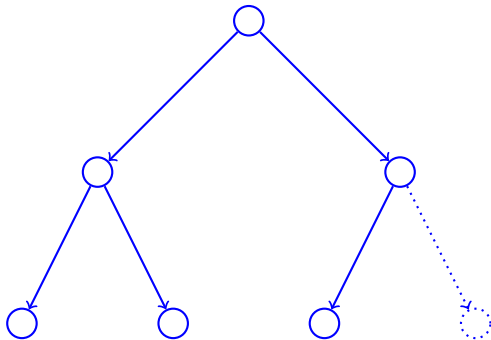
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- ▶ Same argument for d .



Number of nodes

A tree with n arcs has $m = n + 1$ nodes.

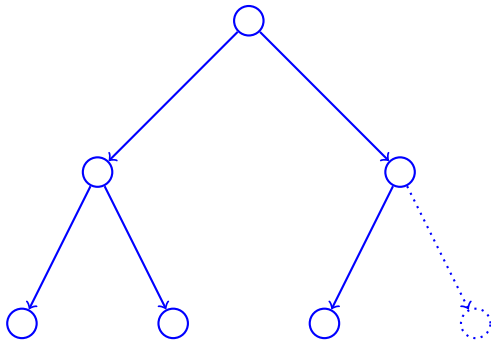
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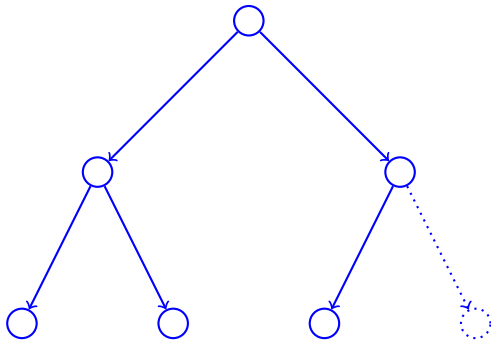
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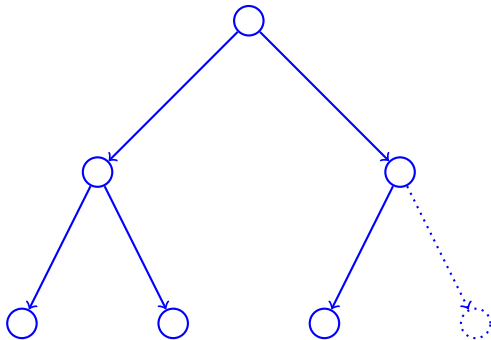
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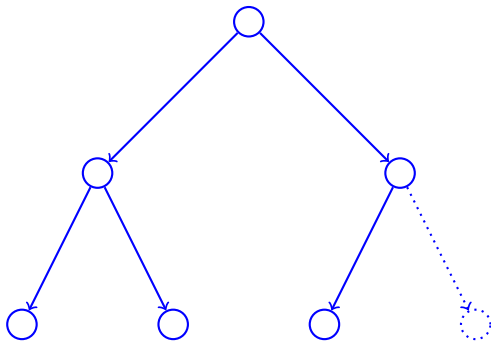
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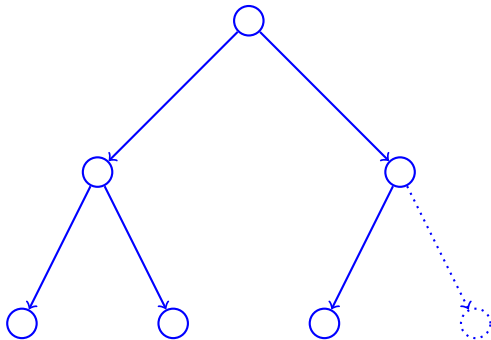
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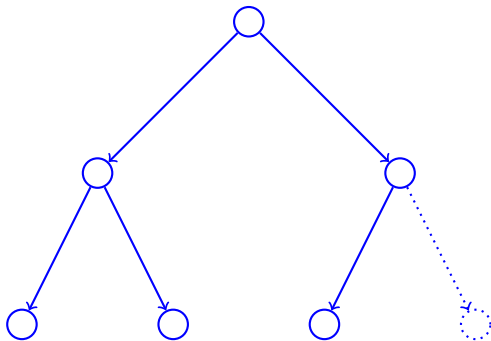
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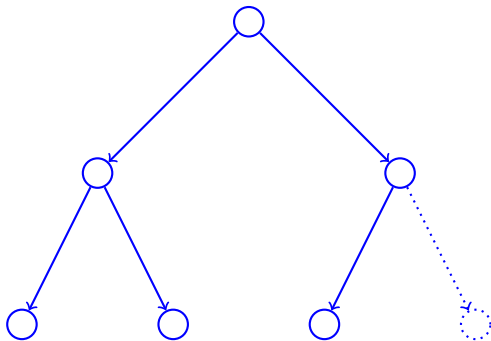
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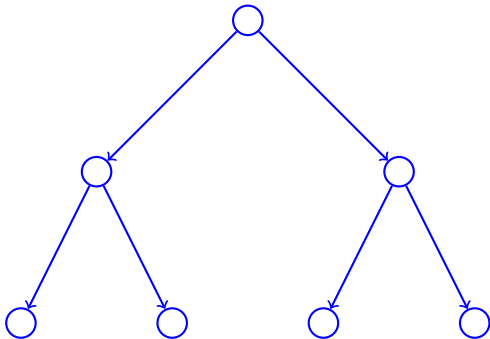
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- ▶ It has p nodes.
- ▶ The original tree has $p + 1$ nodes.



Single path

In a tree, there is exactly one path between any two nodes.

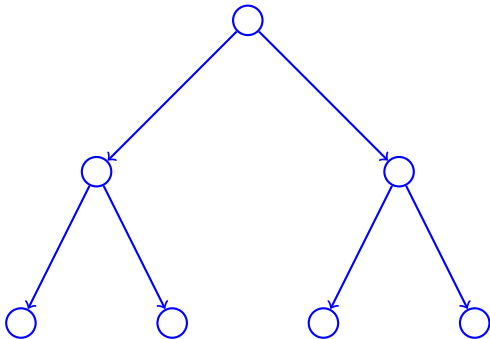
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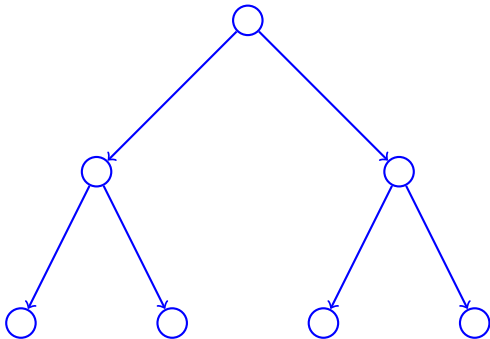
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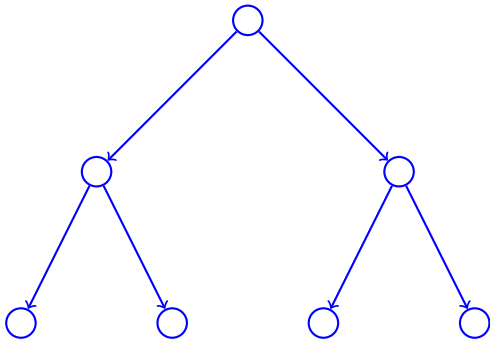
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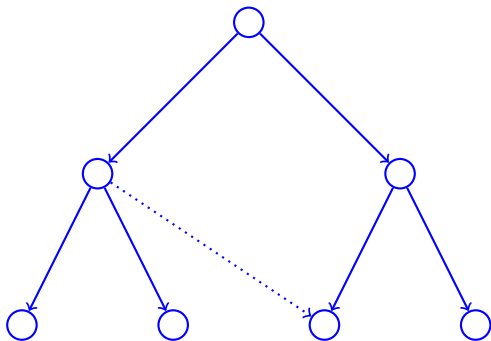
- ▶ A tree is connected, so there is at least one path.
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- ▶ Impossible in a tree.



Cycle formation

In a tree, adding any arc forms a cycle.

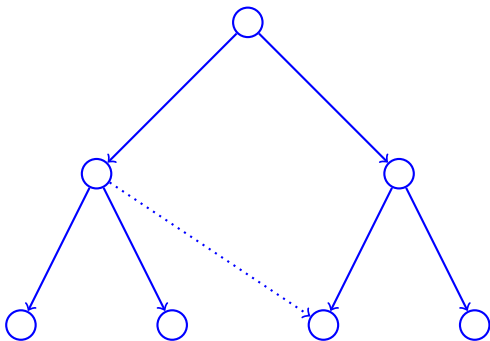
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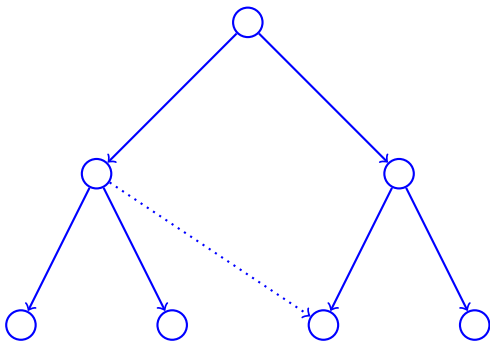
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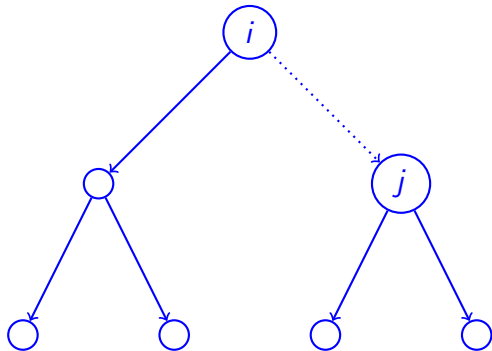
- ▶ Consider adding arc (i, j) .
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- ▶ The added arc forms a cycle with the path.



Disconnection

In a tree, removing any arc disconnects the graph.

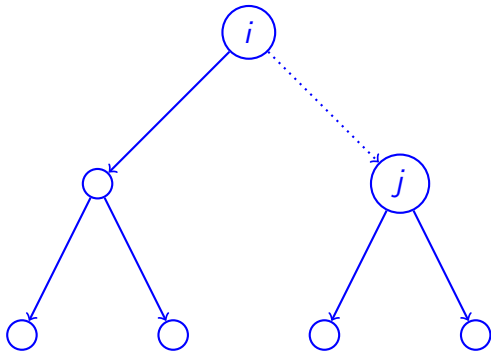
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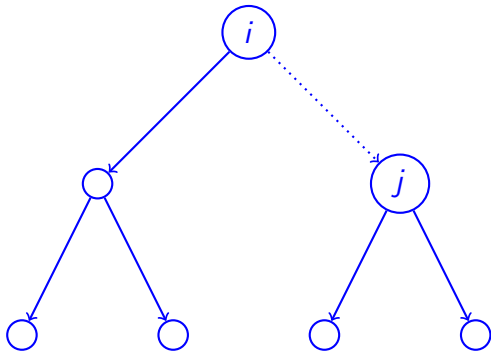
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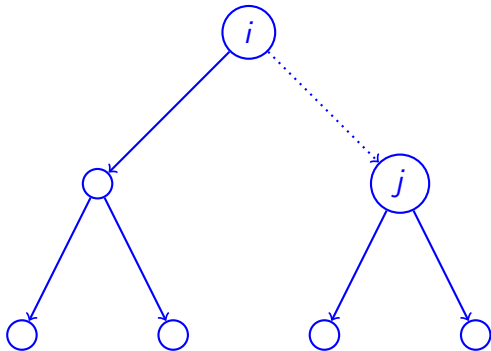
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- ▶ Consider arc (i, j) .
- ▶ There is a unique path from i to j .
- ▶ It is the link!
- ▶ Removing it disconnects i from j .



Characterization

Consider $G = (\mathcal{N}, \mathcal{A}, \phi)$ a directed graph with m nodes and n arcs. The following statements are all equivalent.

- ▶ G is a tree;
- ▶ G is connected and without cycles;
- ▶ There is a unique simple path connecting any two nodes;
- ▶ G has no cycle, and a simple cycle is formed if any arc is added;
- ▶ G is connected and the removal of any single arc disconnects the graph;
- ▶ G is connected and $n = m - 1$;
- ▶ G has no simple cycle and $n = m - 1$.

Summary

We have seen several important properties of trees

They have at least two leafs

The number of nodes is the number of arcs $+ 1$

Each pair of node is connected by exactly one path

Adding an arc forms a cycle

Removing an arc disconnects the graph.